

Deskinator Final Team Report

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Executive Summary

The purpose of this project was to design and prototype the Deskinator, an autonomous robot that cleans classroom desks whenever the user chooses, providing professors and students with a simple way to maintain a clean workspace on demand. Classroom desks accumulate dust, crumbs, and other debris throughout the day, and there is currently no quick, convenient, or hands free method to clean an individual desk without relying on manual wiping. The Deskinator solves this by combining autonomous navigation with a compact vacuum system to deliver fast, user-activated cleaning.

The engineering process included defining clear performance metrics, generating multiple design concepts, and selecting the optimal approach using morph charts, PCC analysis, and early hardware prototyping. The final design uses three line-following sensors to guide the robot along the perimeter of the desk, two DC motors for differential drive, and a central vacuum mechanism that pulls debris toward an internal collection chamber. These components were chosen for their low cost, reliability, and ease of integration into a compact form factor.

Testing showed that the Deskinator can navigate a standard classroom desk reliably, maintain stable detection of the boundary line, and remove small debris such as dust, crumbs, and rice. The device completes a full cleaning pass in approximately 1.4 minutes and achieves strong surface coverage with minimal calibration. While the vacuum struggles with heavier debris and the robot does not reach tight corners perfectly, it successfully meets its main goals and performs effectively for quick, user-activated cleaning.

Overall, the Deskinator demonstrates a practical and reproducible solution for maintaining desk cleanliness whenever needed. Its simple controls, low-cost components, and reliable navigation make it a promising early-stage prototype that can support cleaner classrooms with minimal user effort.

Introduction

Modern classroom environments require quick and convenient ways to maintain clean workspaces, yet most professors and students do not have the time to remove dust, crumbs and debris throughout the day. The Deskinator was developed to address this exact need by providing an autonomous, user activated desk cleaning solution that improves hygiene without having to manually intervene quickly and efficiently.

Problem Statement

The goal is to design a small autonomous robot to safely navigate classroom desks and collect debris, helping professors maintain a clean workspace.

Key Objectives and Metrics (See Appendix A for complete metrics table)

The deskinator was designed under 4 key objectives, each paired with measurable metrics. First, the robot must be 100 percent autonomous, meaning it should be able to complete the cleaning cycle without any external inputs and manual interventions. Second, it must be precise, with a target of cleaning more than 95% of desk debris and trash during operation. Cost was also a key constraint, so the system needed to be cost effective, with a total project cost kept under \$200. Finally, the design had to be safe, incorporating emergency stop features and avoiding any exposed components that could harm the user or the environment. These objectives guided the engineering decisions and served as benchmarks during testing and evaluation.

2. Important or Innovative Design Elements (See Appendix J - Motor Comparison)

When we were thinking about how to fulfill our objectives and metrics, we had to research parts that could move The Deskinator, parts that could detect the edges of the table, and parts that could create suction, and we had to come up with housing that could contain all of these components. We created a morph chart to come up with means that could cover these functions. We focused on movement and detection first, since our housing would have to change to accommodate the components that would move our robot and the components that would detect the table. When we compared motors, we noticed that DC motors would work great for keeping a fan constantly running, which would solve our suction problem, and that stepper motors had great precision, which would have been good for wheels. In our final design, we did not end up going with stepper motors because if we use DC motors with a driver, we can still get good precision and better performance, which is important to fulfill our speed requirements for wheels, and we also ended up using a premade fan instead of using DC motors.

Description of final product

The Deskinator is a compact 12" × 12" autonomous desk-cleaning robot that integrates a simple mechanical, electrical, and software system. (Appendix E). It uses two DC motors for wheel drive, a rear caster for stability, and a centrally mounted vacuum module that collects debris into a removable compartment. Three IR line sensors mounted at the front and sides detect the desk boundary to detect edges (Appendix J). The software runs a basic algorithm where the motors will run until it detects an edge, then it will turn (Appendix G). Two START buttons will initiate the cleaning cycle, one switch to turn it on and one hand wave on the top, after which the robot follows the perimeter of the desk and completes one loop before shutting off. Components were

mounted using 3D-printed brackets, with particular attention to sensor height calibration and motor alignment to ensure stable navigation and consistent suction performance.

Evaluation of results

The Deskinator was able to meet most of its metrics. It was able to meet its first metric of cleaning the desk autonomously, without any inputs. It achieved this by coding a pathing algorithm, the deskinator followed a zig zag pattern along the desk turning when it hit an edge detected by the line followers in the front of the housing (Appendix G). The Deskinator was not able to fully meet its metric of collecting >95 % of the debris, the measured average collection was 81% at the fans maximum speed which is its operating speed, as seen in Appendix J.6. This is due to the fan not producing enough air pressure on the bottom of the collection container. If we had used a stronger/bigger fan this could have improved the product. But our results are still satisfactory. The total cost of the Deskinator (Appendix E) totaled at \$120.96 which is well within the allocated budget of \$200. However, given this budget we could have spent more on better components to enhance the performance of our product. The Deskinator achieved its metric of being safe by being designed with smooth corners and implementing an emergency stop. This emergency stop had a success rate of 97.6% making it suitable for use (Appendix J.5). The Deskinator was able to clean the desk within 2 minutes achieving an average of 84.3 seconds at its operating DC motor PWM signal of 80 which corresponded to around 0.25m/s. Not enough testing was produced to evaluate its durability to meet its metric of 500 cycles without failure; however 50 cycles were tested with no sign of diminishing performance. As seen in Appendix I. Power Budget Calculations the total power consumption of our product is 0.294Wh. Our Power Supply has total power of 307.2Wh allowing the deskinator to run for 87 hours with all of its components at operating voltage and current. Thus, it meets its metric of being to run for more than 5 minutes. The size of the deskinator was measured to be 11 x 9 inches without wheels and 11 x 10.18 inches with wheels meeting its requirement of being less than 12x12 inches. The deskinator was able to successfully indicate completion with a success rate of around 90% meeting its metric. It was not able to attain more because of inconsistency in the dc motors in the turns. The Deskinator was able to start with a touchless signal seen in Appendix J.5 achieving a success rate of 97.6%. Finally the Deskinator was not able to meet its metric of being under 750g and was measured at 1.11 kg.

Lessons learned

We learned that sensor placement is critical; small changes in the height and angle of the IR sensors had a major impact on boundary detection, and early prototypes failed frequently until we calibrated their position. DC motors proved to be the right choice, offering the speed needed for the vacuum and enough control for line-following when paired with three-sensor redundancy. We also found that the debris collection chamber worked well for light particles but needs a larger intake or stronger suction for heavier debris. On the process side, the PCC chart helped us prioritize autonomy and reliability over nonessential features, and early prototyping exposed alignment and calibration issues before final assembly. Future improvements include better corner handling, stronger suction, battery monitoring, and a more efficient cleaning pattern to increase coverage.

References

Terrier GPT – Used to assist with phrasing and wording during report writing.

eSiMotion. *Key Differences Between DC and Stepper Motors*. Available at:

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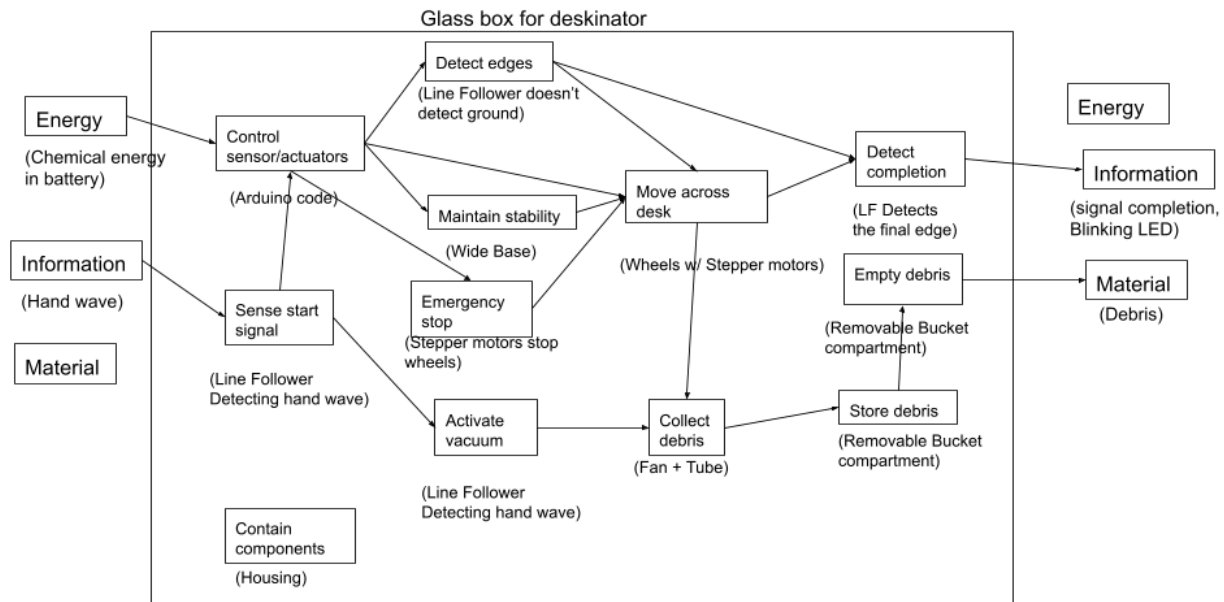
<https://www.youtube.com/watch?v=ygrslqWOh3Y&t=641s>

Appendix

A. Objectives/Metrics Table

Objectives	Metrics
Autonomous	Completes cleaning without external inputs
Precise	Cleans > 95% of desk debris and trash
Cost-effective	Costs < \$200
Safe	Has emergency stops and no exposed sharp objects
Cleans < 2 mins	Has powerful motors that run quickly and cover distance easily
Durable	Cleans through 500 cycles before failure
Runs > 5 mins	Holds enough power to run for 5 minutes
Small	Maximum footprint of 12” x 12”
Indicates completion	Successfully indicates completion of cleaning >90% of the time
Touchless start	Successfully starts cleaning when the desired signal is shown >90% of the time
Light-weight	Weighs <750g

B. Glass Box



C. Pairwise Comparison Chart

Objectives /constraints	Small	Autonomous	Weight	Clean < 2 mins	Touchless start	Cost	Safety	Run >5 mins	Indicate completion	Durability	Precision	Total
Small	—	0	1	0	1	0	0	0	1	0	0	3
Autonomous	1	—	1	1	1	1	1	1	1	1	1	10
Weight	0	0	—	0	0	0	0	0	0	1	0	1
Clean < 2 mins	1	0	1	—	1	0	1	1	1	0	0	6
Touchless start	0	0	1	0	—	0	0	0	0	0	0	1
Cost	1	0	1	1	1	—	1	1	1	1	0	8
Safety	1	0	1	0	1	0	—	1	1	1	0	6
Run >5 mins	1	0	1	0	1	0	0	—	1	0	0	4
Indicate completion	0	0	1	0	1	0	0	0	—	0	0	2

Durability	1	0	0	1	1	0	0	1	1	—	0	5
Precision	1	0	1	1	1	1	1	1	1	1	—	9

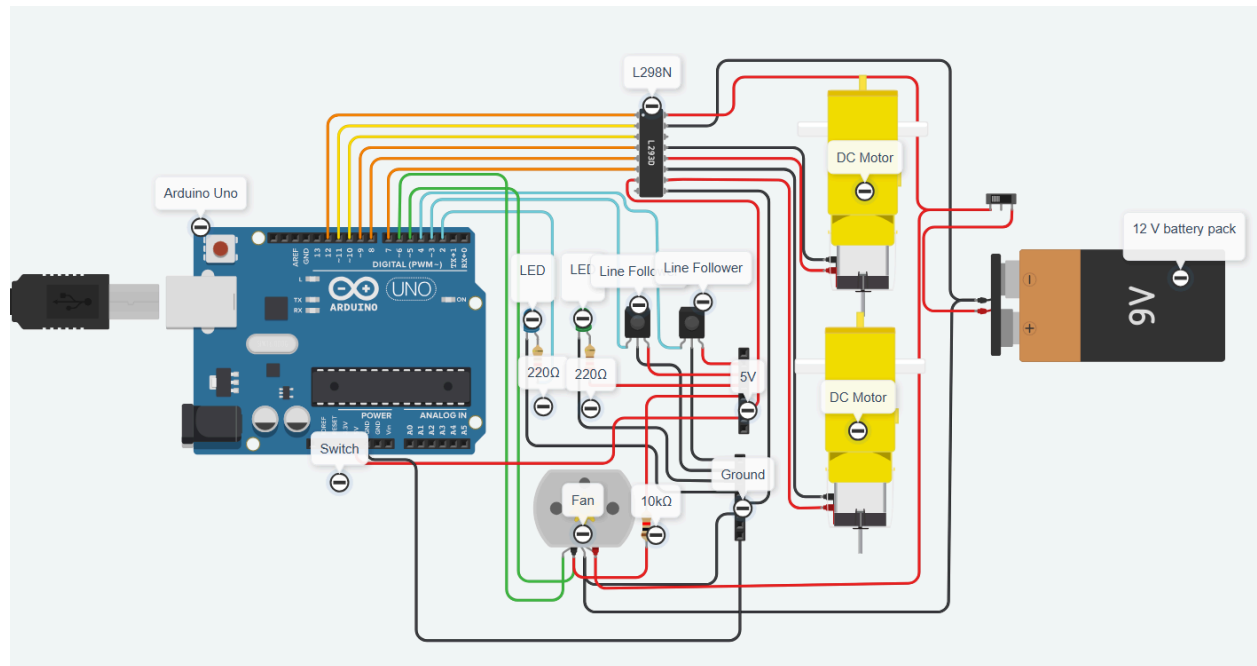
D. Morph Chart

Means Function	1	2	3
Contain Debris	Bucket	Box	Tube
Move Across Desk	Wheels	...	...
Empty container	Twist Cap	Removable Compartment	Openable hatch

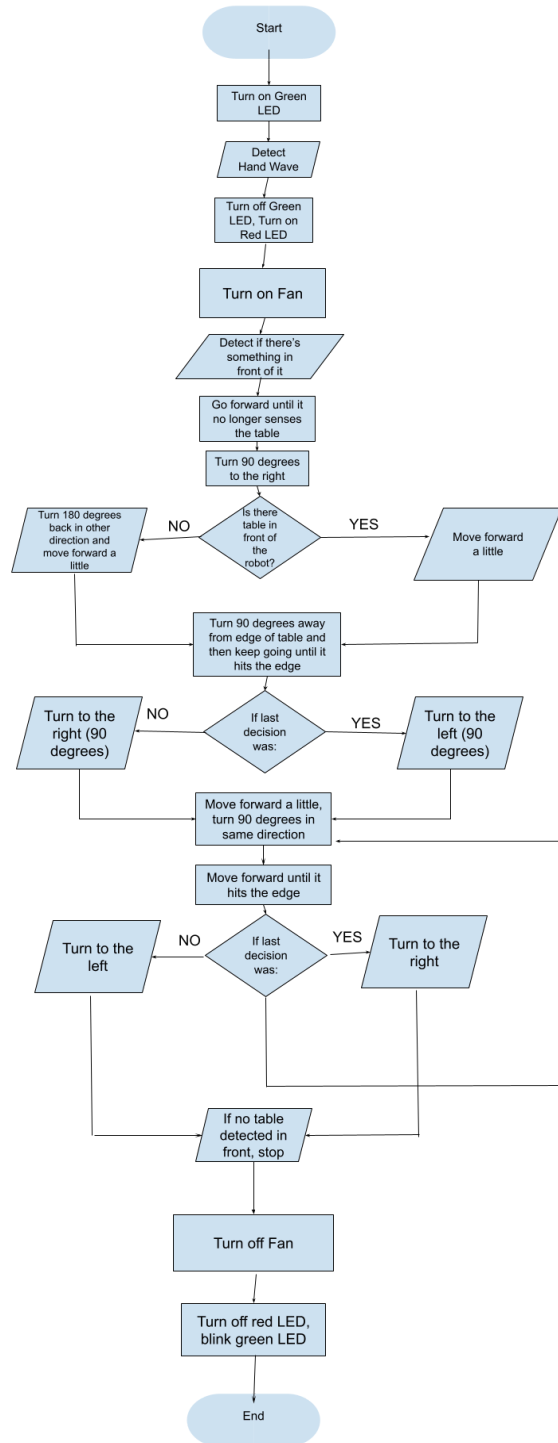
E. Bill of Materials

Part Name	Quantity	Unit cost(\$)	Total Cost(\$)
2 Dc motors + Motor Driver	1	7.99	7.99
Caster wheel	1	1.25	1.25
3D printing	1	20	20
Battery pack	10	10	10
Noctua NF A9 PWM	1	18.95	18.95
3 Line followers	3	0.869	2.607
LED	1	0.70	0.70
Switch	1	3	3
Arduino Uno	1	27.99	27.99
Assembly	N/A	5	5
Total Cost(\$)			120.96

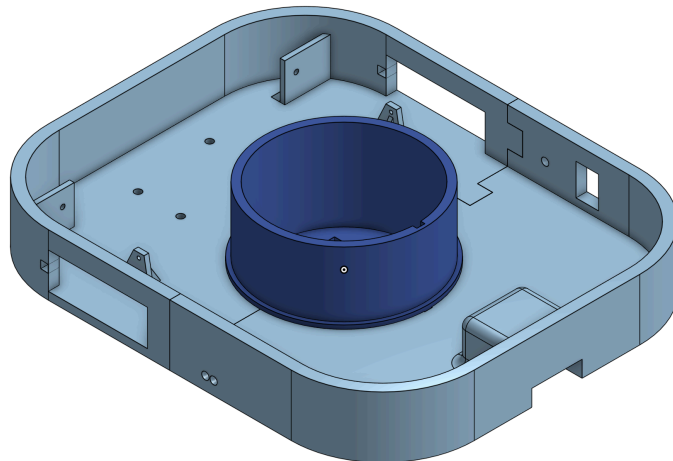
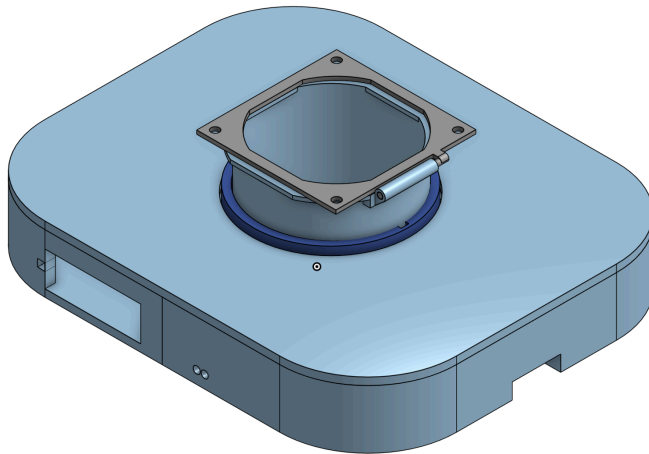
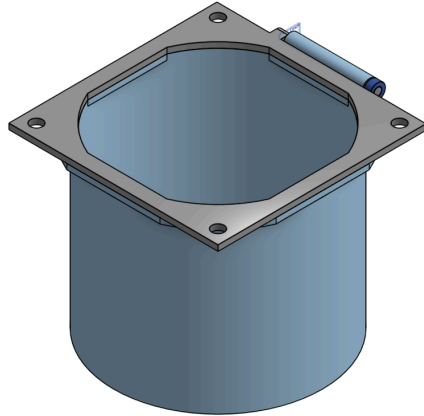
F. Electronics

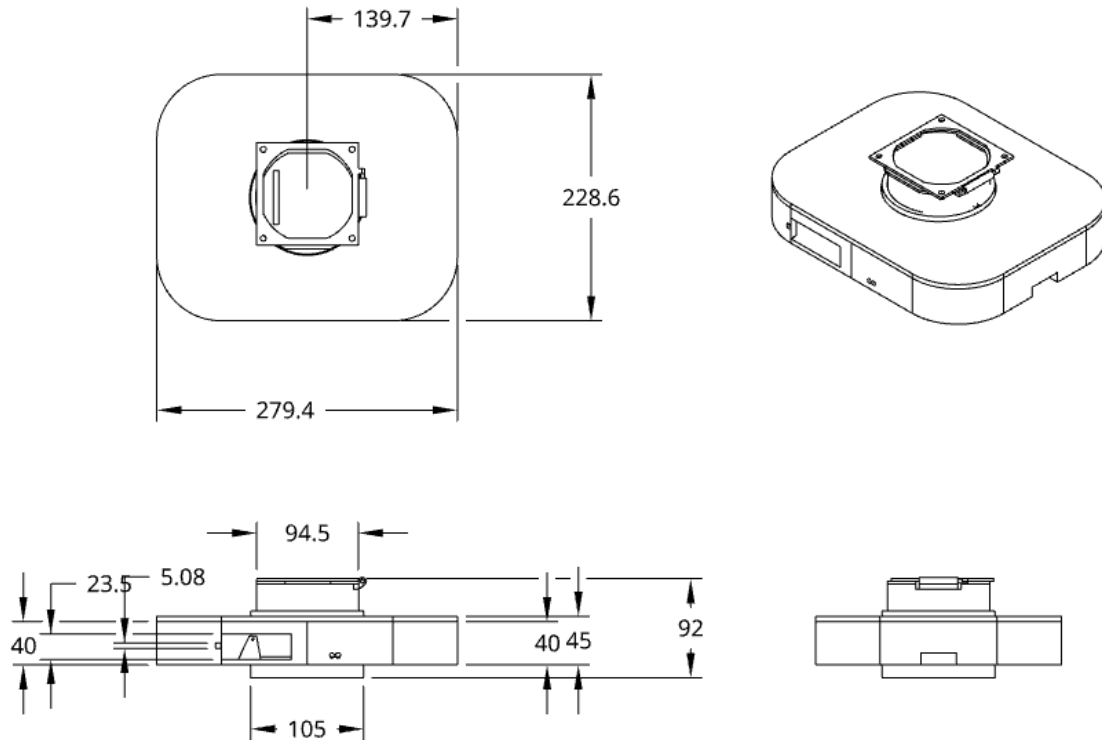


G.1 Code Flow Chart



H. CAD Drawings





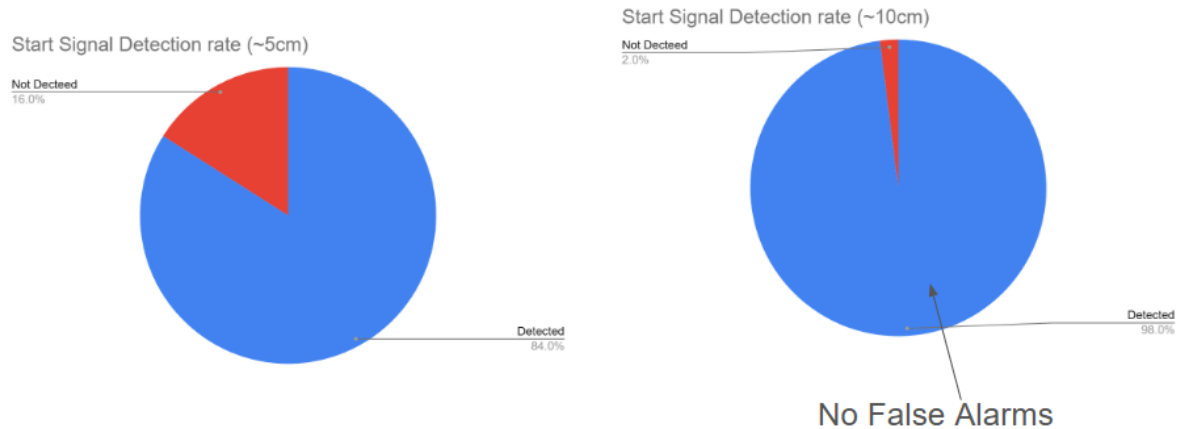
I. Power Budget Calculations

Component	Voltage	Current	Power	Duty Cycle	Total Power
DC motors(2)	10V	300mA	3W	31.4%	0.942W
Noctua nf-a9 PWM	12V	100mA	1.2W	100%	1.2W
L298N	12V	36mA	0.432W	100%	0.432W
Line followers	5V	15mA	0.25W	100%	0.25W
Green LED	5V	9mA	0.45W	100%	0.45W
Arduino Uno	5V	50mA	0.25W	100%	0.25W
Total					3.524
				Total time(mins	5

				Energy Used	0.294Wh
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J. Additional plots and tables

J.1 Edge Detection Results



Edge Detection Rate = 100%

J.2 Motor Comparison Table

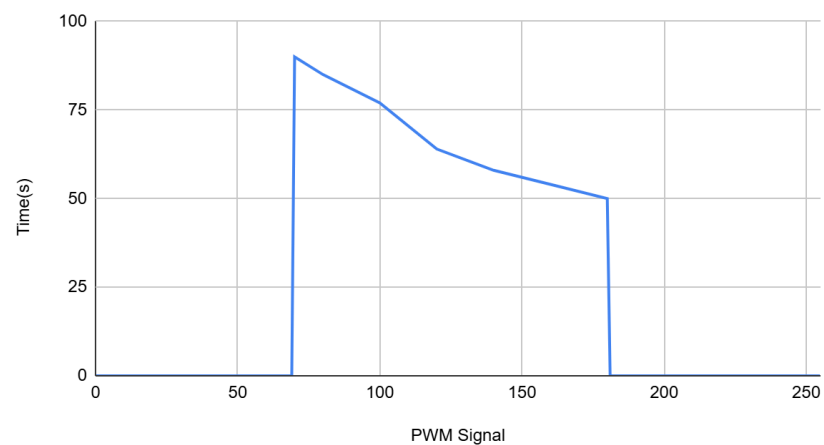
	DC	Stepper (28BYJ-48)
Speed	~10,000 RPM @ 6V	~15 RPM (full step)
Torque	~0.10 kg·cm	~0.34 kg·cm
Precision	Continuous Rotation	2048 steps/rev
Usefulness	Great for fan	Great for wheels

J.3 Line Follower Detection Table

Potentiometer Position	Detection Length	Picture
0 tick marks	Always off	
1 tick mark	1.5 in	
2 tick marks	5 in	
3 tick marks	15.75 in	
4 tick marks	Always on	

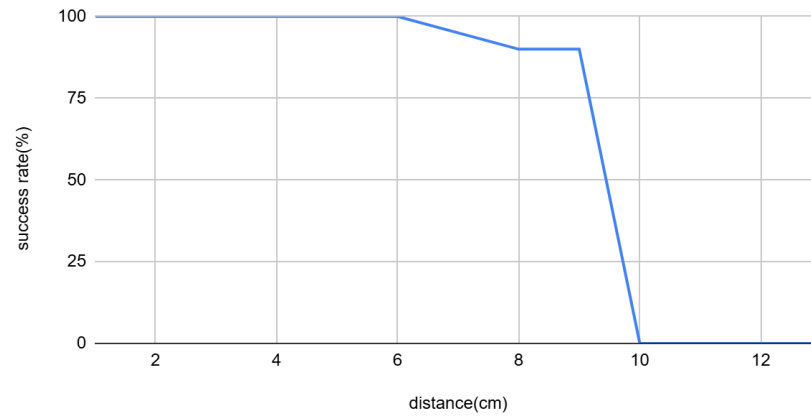
J.4 Time of Completion vs. Fan Speed plot

Time(s) of completion vs. PWM Signal of DC motors



J.5 Signal detection rate of Line followers vs Distance plot

success rate of start and emergency signal(%) vs. distance from sensor(cm)



J.6 Debris collected vs Speed of DC motor plot

% Debris vs. PWM Signal

